Cutting the Cord: Authority-Independent, Escrow-Free Authentication and Key Management for TUAV-Assisted Infrastructure-Less IoV

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***Abstract*—A recent scheme by Tan, Zheng and Vijayakumar (IEEE T-ITS 2023) uses a tethered UAV (TUAV) to replace roadside units destroyed by a disaster, so that an Internet of Vehicles keeps working "without infrastructure". We show that it does not achieve this. Every authentication is relayed over a satellite link to a cloud Trusted Authority (TA) that computes all pairings. The UAV credential depends only on values the TA holds, so the TA can impersonate any UAV. The CRT-based group key reveals every member's decrypting key to every other member, so a revoked UAV still obtains the new key. The reported costs omit the TA's pairings and an $O(n^3)$-byte key broadcast. We confirm these findings in code and, for the escrow attack, with ProVerif. We then propose *Cutting the Cord*, which moves the trust root offline. The TUAV authenticates UAVs itself with escrow-free, pairing-free certificateless signatures in a sound batch with a bounded fallback. It delivers the group key per member under fresh ECDH keys, admits vehicles through authenticated UAV relays, and hands members between TUAVs with single-use tickets. ProVerif verifies authentication, key secrecy, forward secrecy and revocation, and delimits the residual TA power to Girault level 2. On the same curve and library, the base scheme's TA-inclusive cost is about 65× ours, our scheme needs no satellite traffic, and it sends 1,823× less key traffic.**

***Index Terms*—Internet of Vehicles, tethered UAV, certificateless signatures, batch verification, group key management, ProVerif.**

I. Introduction

Roadside units (RSUs) authenticate vehicles in a conventional Internet of Vehicles (IoV). After an earthquake or flood they are among the first things to fail, exactly when rescue traffic needs secure coordination. Tan *et al.* [1] propose a *tethered UAV* (TUAV), powered by cable from an emergency vehicle, as a flying edge server for a swarm of UAVs, with certificateless batch authentication and a CRT-based group key. They report 151 ms to authenticate 120 UAVs and eleven satisfied security properties.

We examined the scheme equation by equation, implemented it on BN254, and modelled its credential in ProVerif. The architecture is sound; the protocol is not. The TUAV cannot verify anyone without a satellite round trip to the TA (so authentication stops when the link fails). The TA can impersonate every UAV. A revoked UAV keeps receiving group keys. The efficiency figures omit the dominant cost.

Contributions. (1) Six findings against [1], four reproduced in code and one found automatically by ProVerif (Sec. III). (2) *Cutting the Cord*: a framework with an offline trust root and three tiers. C1 is autonomous, escrow-free UAV authentication with secure rekeying. C2 admits vehicles through UAV relays. C3 is ticket-based handover between TUAVs (Sec. IV). (3) Security arguments and five ProVerif models with sixteen queries, which pin down the residual TA power precisely (Sec. V). (4) An implementation of both schemes on one curve and library, with eight experiments (Sec. VI).

II. Background and the Base Scheme

Certificateless cryptography [2] splits a private key into a partial key issued by the authority and a secret value the authority never sees. In Girault's terms [3], the authority is then at *level 2*: it cannot learn a user's key, but it could issue a second, detectable key. At level 1 it knows users' keys. **Batch verification** needs random small exponents to be sound [4]. **Group key management** ranges from per-member (unicast) delivery to key trees (LKH) [6] and single-broadcast CRT "secure locks" [7].

The base scheme [1] has a TA, a TUAV and n UAVs. In IS1–IS5 the TA gives the TUAV (id_{tu}, s_{tu}) and each UAV (id_i, s_i), and the TUAV broadcasts $(ts_{tu}, ID_{tu}, R_{tu}, Q_{tu})$. In US1–US3 each UAV picks r_i and sends $(ts_2, ID_i, \xi_i, \upsilon_i, \Gamma_i)$, where $\xi_i = h_1(r_i)$ and

$$\Gamma_i = [s_i h_3(ID_i, ts_2, s_i) R_{tu} - h_1(s_i \xi_i)]P - \upsilon_i h_1(r_i) Q_{tu}. \quad (1)$$

In GA1–GA2 the TUAV uploads the requests to the TA, which computes $\sigma_i = h_1(s_i \xi_i)P$ and three pairings ψ_i, μ_i, δ_i. In GA3 the TUAV checks one product equation. In GD1–GD6 the TUAV publishes the coefficients of

$$\Lambda(x) = \kappa_{tu} \textstyle\sum_i \omega_i \varphi_i + \prod_i (x - \sigma_i), \quad (2)$$

with $\omega_i = \prod_{j \neq i} \sigma_j$ and $\varphi_i \equiv \omega_i^{-1} \bmod \sigma_i$. Each UAV recovers $\kappa_{tu} = \Lambda(\sigma_i) \bmod \sigma_i$. In DU1 the polynomial is rebuilt over the remaining members' *unchanged* σ_i.

III. Analysis of the Base Scheme

F1: not infrastructure-less. GA3 uses only values computed by the TA in GA2, and the TA is reached by satellite. With link availability A, a round succeeds with probability A, and every round pays a satellite round trip.

F2: the TA impersonates every UAV. Since $\xi_i = h_1(r_i)$ is sent in clear, (1) depends only on s_i, which the TA stores, and on public values. The TA can therefore build a valid request for any UAV (level 1, so claims F4 and F10 of [1] fail).

Verification also needs s_i, so no credential is non-repudiable. Our implementation accepts a TA-built request, and ProVerif finds the attack (Q16, Table I).

F3: no forward secrecy. σ_i is determined by s_i and the public ξ_i; a later leak of s_i opens every recorded group key.

F4: revocation fails.

Lemma 1. *For every member j: $\Lambda(x) - \Lambda(\sigma_j) = \prod_i (x - \sigma_i)$.*

Proof. $\Lambda(x) = C + \prod_i (x - \sigma_i)$ with constant C, and the product vanishes at σ_j. □

Any member therefore recovers *all* σ_i by integer root finding. Because DU1 reuses them, a revoked UAV computes the new key as $\Lambda^*(\sigma_j) \bmod \sigma_j$ for any staying j. Our code executes this attack. The σ_i are also points in GA2 but moduli in GD1, so the construction is not even well defined as printed.

F5: empty proof, unsound batch. Theorem 1 of [1] reduces forgery to the CDH instance $(s_{tu}P, (\sum v_i \xi_i)P)$, whose second exponent is public, so the instance is trivial. GA3 has no random weights, so $\Gamma_1 + \Delta, \Gamma_2 - \Delta$ pass while both fail individually (reproduced). GA4 has no failure branch, so one bad request rejects everyone.

F6: cost omits the TA. GA2 performs three pairings per UAV. With the base paper's own MIRACL constants, group authentication of 120 UAVs costs 3720 ms plus the satellite RTT, not 151 ms. The CRT coefficients make each acknowledgement $O(n^2)$ bits, sent to each of n UAVs (Sec. VI).

All six share one root cause: *the TA is inside every operation*.

IV. Cutting the Cord

Model. A TA (offline after deployment); TUAVs $\mathcal{T}_a$ with public identities; UAVs $\mathcal{U}_i$ and vehicles $\mathcal{V}_j$ with pseudonyms. The attacker is Dolev–Yao on all links, plus Type-I (key-replacing outsider) and Type-II (malicious TA) certificateless adversaries, insiders, and batch poisoners.

Setup and registration (offline). The TA holds s ($P_{\text{pub}} = sP$) to certify keys and t ($T_{\text{pub}} = tP$) to trace pseudonyms. For each epoch ep an entity picks x, sends $X = xP$ with a proof of possession, and receives $PID = (\rho P, RID \oplus H_0(\rho T_{\text{pub}} \,\|\, ep))$, $Y = yP$ and $d = y + sh$, where $h = H_1(PID, ep, X, Y)$. The full key is $sk = x + d$ and $pk = X + Y + hP_{\text{pub}} = skP$. The TA knows d but never x.

C1, UAV tier. The TUAV broadcasts a signed beacon carrying a fresh $E_a = e_a P$. A UAV signs $m_i = (\texttt{join} \,\|\, ID_a \,\|\, E_a \,\|\, E_i)$ with a Schnorr-type signature (R_i, v_i), $v_i = r_i + c_i sk_i$ for challenge c_i, and sends $\langle PID_i, ep, X_i, Y_i, E_i, ts_i, R_i, v_i \rangle$. The TUAV filters the requests (freshness, epoch, eviction list, one per pseudonym) and checks, with random 64-bit λ_i,

$$\Big(\sum \lambda_i v_i\Big)P = \sum \lambda_i R_i + \sum (\lambda_i c_i)(X_i + Y_i) + \Big(\sum \lambda_i c_i h_i\Big)P_{\text{pub}}. \tag{3}$$

If the check fails, the TUAV verifies each request, admits the valid ones, evicts the bad pseudonyms for the epoch, and stays in individual mode until windows are clean. It then derives $K_i = \mathsf{KDF}(e_a E_i \,\|\, \dots)$ and broadcasts one ACK with $\mathsf{Enc}(K_i, GK_a \,\|\, \nu)$ for each member. Rekeying draws a fresh GK'_a and encrypts it only under the remaining K_i; LKH [6] reduces a single leave to $2\lceil \log_2 n \rceil - 1$ ciphertexts.

C2, vehicle tier. Vehicles hold the same credentials. A UAV relays their signed requests in a bundle authenticated by $\mathsf{MAC}(K_i, \cdot)$, so the TUAV discards bundles from non-members before any signature work. Valid vehicles receive a separate regional key VK_a. Routine V2V beacons carry a MAC under VK_a; safety-critical events are signed.

C3, handover. Neighbouring TUAVs agree a link key K_{ab} by signed ECDH and share eviction lists. Before a member $\mathcal{M}$ leaves, $\mathcal{T}_a$ gives it a ticket $\mathsf{Enc}(K_{ab}, PID \,\|\, pk \,\|\, hk \,\|\, N \,\|\, exp)$, where $hk = \mathsf{KDF}(K_{\mathcal{M}} \,\|\, \dots)$. $\mathcal{T}_b$ opens the ticket, rejects reused N, verifies a MAC under hk, runs one fresh ECDH, and sends its group key. No signature is verified, and $\mathcal{T}_b$ never learns $K_{\mathcal{M}}$.

Tracing and revocation. Offline, the TA recovers $RID = PID^2 \oplus H_0(tPID^1 \,\|\, ep)$ and can publish the remaining pseudonyms of a revoked entity. TUAVs evict immediately and locally, so nothing waits for the TA.

V. Security Analysis

Proposition 1. *No step of C1–C3 sends to or waits for the TA; verification needs only P_{pub}.*

Theorem 1 (Unforgeability)**.** *Under ECDLP in the ROM, no Type-I or Type-II adversary forges a signature under an honest entity's registered key.*

Sketch. Type II: embed the challenge in X^*; forking on H_2 yields sk^*, hence $x^* = sk^* - d^*$. Type I: embed it in P_{pub}; double forking on H_2 and on H_1 yields $(sk^* - sk^{*\prime})/(h^* - h^{*\prime})$ [8]. Unlike [1], the challenge is never a public value. □

Theorem 2 (Batch soundness [4])**.** *If any request is invalid, (3) holds with probability at most 2^{-64}.*

Theorem 3 (Keys)**.** *Under CDH and AEAD security, K_i is known only to $\mathcal{T}_a$ and $\mathcal{U}_i$. Session keys survive later compromise of all long-term keys. A removed member cannot obtain GK'_a, and a joiner cannot obtain earlier keys.*

Proposition 2 (Bounded poisoning)**.** *With $d \geq 1$ invalid requests, a window costs at most $T_{BV}(n) + nT_{SV}$, and every valid request is admitted.*

ProVerif [9]**.** Table I lists the sixteen queries. Q7 and Q9 fail *as predicted*. A malicious TA can still issue a second key for a pseudonym, or enrol a phantom member and receive the group key. This is inherent to certificateless schemes and detectable. Q6 and Q8 show that it cannot sign under the registered key or read pairwise traffic. Q12 holds only non-injectively, because ProVerif's table lookup and insert are not atomic. Our TUAV processes handovers sequentially, and a replayed ticket still yields no key (Q13–Q15).

TABLE I
PROVERIF 2.05 RESULTS (ALL AS PREDICTED).

#	Property	Result
Q1–Q3	C1 mutual authentication (Type I, injective)	✓ true
Q4–Q5	group-key secrecy; forward secrecy (all keys leaked)	✓ true
Q6	malicious TA cannot use the *registered* key	✓ true
Q7	malicious TA with a *new* key (key replacement)	false (level 2)
Q8	pairwise-key data secret from the TA	✓ true
Q9	group key vs. phantom member enrolled by the TA	false (level 2)
Q10–Q11	revoked member: old key known, new key secret	✓ true
Q12	handover acceptance ⇒ member started it	✓ non-inj.
Q13–Q15	handover completion (inj.), secrecy, forward secrecy	✓ true
Q16	**base scheme**: acceptance ⇒ UAV sent	✗ attack

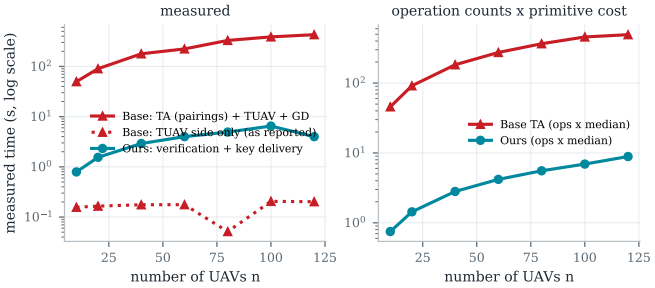


Fig. 1. E2: time to authenticate and key n UAVs (log scale). Left: measured. Right: operation counts × median costs.

VI. Implementation and Evaluation

Setup. Both schemes run on BN254 with `py_ecc` (pure Python) and `cryptography`, in one process on one laptop. The base scheme's symmetric pairing is realised by placing the argument the TA can compute in $\mathbb{G}_2$. For each n the two schemes run back to back. Because the laptop's speed drifted during the day, we also report a load-independent ratio from the exact operation counts recorded by the code. Here a pairing costs about 44 scalar multiplications, against about 8 with MIRACL. Twenty tests pass, covering correctness and every attack.

Latency (E2). For $n = 100$ our TUAV authenticates and keys the swarm in 6.49 s; the base scheme needs 388 s before any satellite delay, and its TUAV-only part (the part [1] reports) is 0.05% of that. By operation count the base scheme does 55–66× more work at every n; measured ratios ($n \le 100$) are 56–67×. With the base paper's MIRACL constants the figures are 228 ms against 3720 ms plus RTT (Fig. 1).

Availability (E1). The base scheme succeeds in a fraction A of rounds; with three retries at $A = 50\%$ it reaches 87.5% under independent losses but only 59.2% under bursty outages. Ours does not use the link.

Poisoning (E3). One invalid request blocks every honest UAV in the base scheme. Ours admits all honest UAVs at every attack rate, with the cost bounded by a failed batch plus individual checks (2.00 s clean, 7.00 s worst). Split-and-retest [5] brings no gain because pairing-free batching is only a few times cheaper per signature (Fig. 2).

Traffic (E4). For $n = 120$ the base scheme's acknowledgements carry 17.5 MB of CRT coefficients and it moves

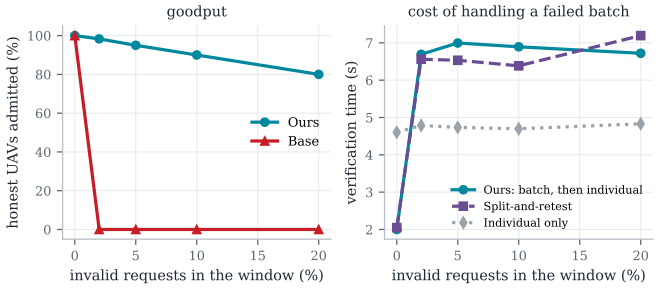


Fig. 2. E3: honest UAVs admitted and verification time under batch poisoning ($n = 60$).

158.6 KB over the satellite link. Our single ACK is 9.6 KB (1,823× less). Our join request is larger (225 B against 137 B).

Vehicles and handover (E5, E6). A bundle of 500 vehicles is admitted, including keys, in 29.6 s, and non-member bundles are dropped in microseconds. A ticket handover costs the new TUAV 52.3 ms instead of 105.0 ms. Over a one-hour mobility run with 557 cell crossings, the mean service gap falls from 756 ms to 103 ms.

VII. Limitations

A malicious TA keeps Girault level-2 power (detectable key replacement and phantom enrolment). A captured TUAV exposes its own cell's keys until it is evicted. Pseudonyms are linkable within an epoch. Requests are larger. Our timings come from a pure-Python library on a laptop, and network effects are simulated. Threshold TUAV trust, manufacturer-bound keys (level 3) and an embedded constant-time implementation are future work.

VIII. Conclusion

The TUAV architecture of [1] is a good answer to infrastructure loss, but its protocol keeps the cloud authority inside every step. Moving the trust root offline restores infrastructure independence, escrow-freeness, forward secrecy and secure revocation together, reduces total work and traffic, and extends naturally to vehicles and to multi-TUAV handover.

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